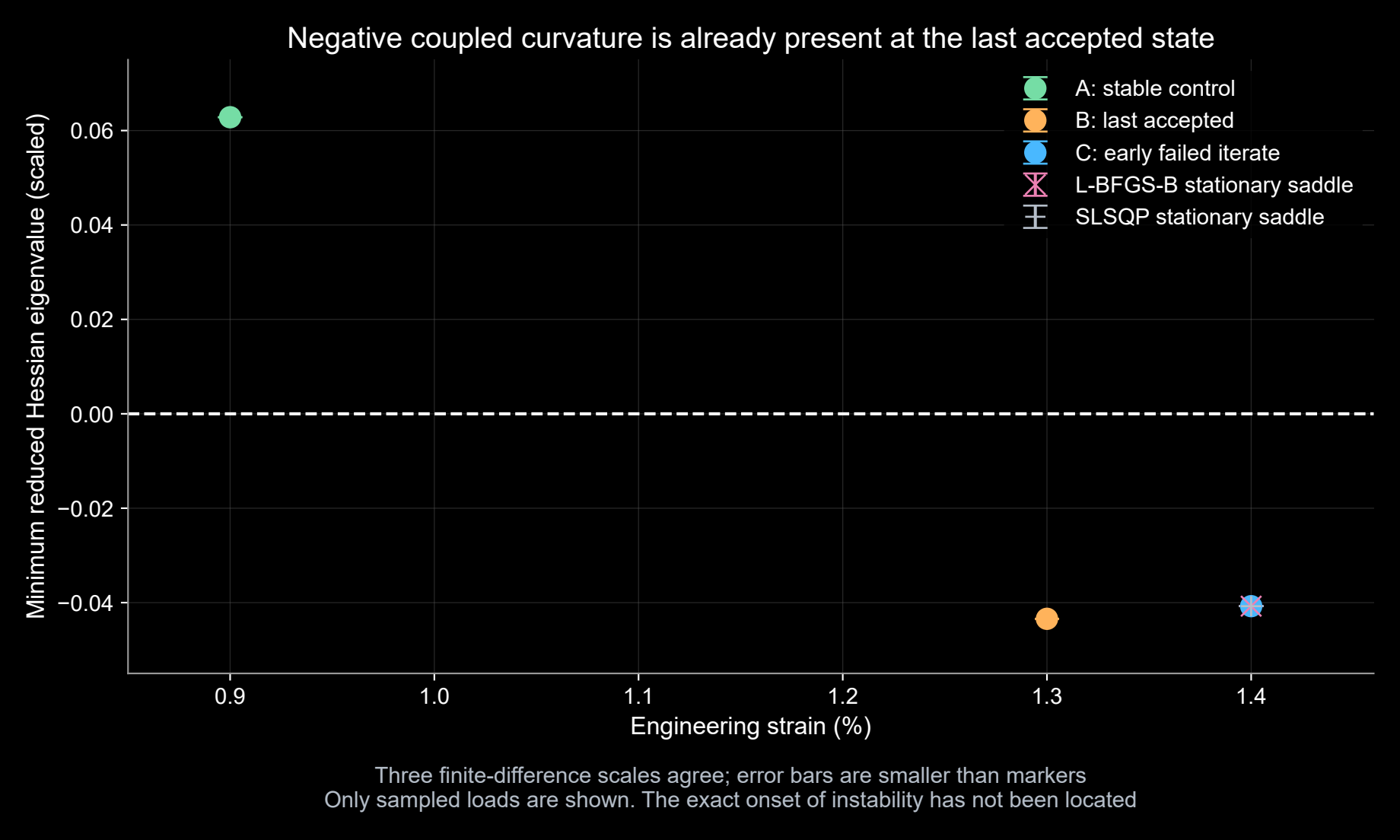
Negative coupled curvature is already present at the last accepted state
A: stable control
B: last accepted
C: early failed iterate
L-BFGS-B stationary saddle
SLSQP stationary saddle
Minimum reduced Hessian eigenvalue (scaled)
0.06
0.04
0.02
0.00
−0.02
−0.04
0.9
1.0
1.1
1.2
1.3
1.4
Engineering strain (%)
Three finite-difference scales agree; error bars are smaller than markers
Only sampled loads are shown. The exact onset of instability has not been located